

Q1Array Index Traversal
5 marks**Q2**Null Pointer Handling
5 marks**Q3**String Comparison
5 marks**Q4**While Loop Correction
5 marks**Q5**Reverse String Logic
5 marks**Q6**Array Sum Calculation
5 marks**Q7**String Character Access
5 marks**Q8**Case Insensitive Comparison
5 marks**Q9**

The following Java program is intended to print all elements of an array. However, it throws an error during execution. Identify and correct the mistake in the loop condition.

```
public class Q1_ArrayIndex {  
    public static void main(String[] args) {  
        int arr[] = {10, 20, 30};  
        for(int i = 0; i < arr.length; i++){  
            System.out.println(arr[i]);  
        }  
    }  
}
```

Your Code

```
1 import java.util.*;  
2 public class Q1_ArrayIndex {  
3     public static void main(String[] args) {  
4         int arr[] = {10, 20, 30};  
5         for (int i = 0; i < arr.length; i++)  
6         {  
7             System.out.println(arr[i]);  
8         }  
9     }  
10 }
```

Input

stdin input

Output10
20
30**Visible Test Cases**

Q1

Array Index Traversal
5 marks

Q2

Null Pointer Handling
5 marks

Q3

String Comparison
5 marks

Q4

While Loop Correction
5 marks

Q5

Reverse String Logic
5 marks

Q6

Array Sum Calculation
5 marks

Q7

String Character Access
5 marks

Q8

Case Insensitive Comparison
5 marks

A program attempts to print the length of a string but results in a runtime exception. Analyze the issue and modify the code to handle it properly.

```
public class Q2_NullPointer {  
    public static void main(String[] args) {  
        String str = null;  
        if(str != null)  
            System.out.println(str.length());  
        else  
            System.out.println("String is null");  
    }  
}
```

Your Code

```
1 public class Q2_NullPointer {  
2     public static void main(String[] args) {  
3         String str = null;  
4         if (str != null)  
5             System.out.println(str.length());  
6         else  
7             System.out.println("String is null");  
8     }  
9 }
```

Input

stdin input

Output

String is null

QUESTIONS

Q1

Array Index Traversal
5 marks

Q2

Null Pointer Handling
5 marks

Q3

String Comparison
5 marks

Q4

While Loop Correction
5 marks

Q5

Reverse String Logic
5 marks

Q6

Array Sum Calculation
5 marks

Q7

String Character Access
5 marks

Q8

Case Insensitive Comparison
5 marks

Q3 String Comparison

5 marks

A developer compares two strings using the == operator, which leads to incorrect output. Identify the issue and correct the comparison method.

```
public class Q3_StringCompare {  
    public static void main(String[] args) {  
        String s1 = "Java";  
        String s2 = new String("Java");  
        if(s1.equals(s2)){  
            System.out.println("Equal");  
        }  
    }  
}
```

Your Code

```
1 public class Q3_StringCompare {  
2     public static void main(String[] args) {  
3         String s1="SIMATS";  
4         String s2=new String("SIMATS");  
5         if (s1.equals(s2)) {  
6             System.out.println("Equal");  
7         }  
8     }  
9 }
```

Input

stdin input

Output

Equal

Q1Array Index Traversal
5 marks**Q2**Null Pointer Handling
5 marks**Q3**String Comparison
5 marks**Q4**While Loop Correction
5 marks**Q5**Reverse String Logic
5 marks**Q6**Array Sum Calculation
5 marks**Q7**String Character Access
5 marks**Q8**Case Insensitive Comparison
5 marks**Q9**

The given program uses a loop to print numbers from 0 to 4, but it runs infinitely. Identify the cause and fix the loop.

```
public class Q4_Loop {  
    public static void main(String[] args) {  
        int i = 0;  
        while(i < 5){  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

Your Code

```
1 public class Q4_Loop {  
2     public static void main(String[] args) {  
3         int i = 0;  
4         while (i < 5) {  
5             System.out.println(i);  
6             i++;  
7         }  
8     }  
9 }
```

Input

stdin input

Output0
1
2
3
4**Visible Test Cases**

Q1

Array Index Traversal
5 marks

Q2

Null Pointer Handling
5 marks

Q3

String Comparison
5 marks

Q4

While Loop Correction
5 marks

Q5

Reverse String Logic
5 marks

Q6

Array Sum Calculation
5 marks

Q7

String Character Access
5 marks

Q8

Case Insensitive Comparison
5 marks

Q9



A program is written to reverse a string. Verify whether the logic is correct and modify it if necessary.

```
public class Q5_Reverse {  
    public static void main(String[] args) {  
        String str = "Hello";  
        String rev = "";  
        for(int i = 0; i < str.length(); i++){  
            rev = str.charAt(i) + rev;  
        }  
        System.out.println(rev);  
    }  
}
```

Your Code

```
1 public class Reverse {  
2     public static void main(String[] args) {  
3         String str = "SIMATS";  
4         String rev = "";  
5         for (int i = 0; i < str.length(); i++) {  
6             rev = str.charAt(i) + rev;  
7         }  
8         system.out.println(rev);  
9     }  
10 }
```

Input

stdin input

Output

STAMIS

Q1

Array Index Traversal

5 marks



Q2

Null Pointer Handling

5 marks



Q3

String Comparison

5 marks



Q4

While Loop Correction

5 marks



Q5

Reverse String Logic

5 marks



Q6

Array Sum Calculation

5 marks



Q7

String Character Access

5 marks



Q8

Case Insensitive Comparison

5 marks



Q9



The program calculates the sum of array elements, but it produces an incorrect result. Identify the problem and fix it.

```
public class Q6_ArraySum {  
    public static void main(String[] args) {  
        int arr[] = {1,2,3,4};  
        int sum = 0;  
        for(int i=0;i<arr.length;i++){  
            sum += arr[i];  
        }  
        System.out.println(sum);  
    }  
}
```

Your Code

```
1 public class ArraySum {  
2     public static void main(String[] args) {  
3         int[] arr = {1, 2, 3, 4};  
4         int sum = 0;  
5         for (int i = 0; i < arr.length; i++) {  
6             sum += arr[i];  
7         }  
8         System.out.println(sum);  
9     }  
10 }
```

Input

stdin input

Output

10

QUESTIONS

Q1

Array Index Traversal
5 marks

Q2

Null Pointer Handling
5 marks

Q3

String Comparison
5 marks

Q4

While Loop Correction
5 marks

Q5

Reverse String Logic
5 marks

Q6

Array Sum Calculation
5 marks

Q7

String Character Access
5 marks

Q7 String Character Access

5 marks

A program tries to access a character from a string using an invalid index, causing an exception. Identify and correct the issue.

```
public class Q7_StringIndex {  
    public static void main(String[] args) {  
        String str = "Java";  
        System.out.println(str.charAt(3));  
    }  
}
```

Your Code

```
1 public class Q7_StringIndex {  
2     public static void main(String[] args) {  
3         String str = "Java";  
4         System.out.println(str.charAt(2));  
5     }  
6 }
```

Input

stdin input

Output

v

QUESTIONS

Q1

Array Index Traversal

5 marks

**Q2**

Null Pointer Handling

5 marks

**Q3**

String Comparison

5 marks

**Q4**

While Loop Correction

5 marks

**Q5**

Reverse String Logic

5 marks

**Q6**

Array Sum Calculation

5 marks

**Q7**

String Character Access

5 marks

**Q8**

Case Insensitive Comparison

5 marks

**Q8 Case Insensitive Comparison**

5 marks

A string comparison fails due to case sensitivity. Modify the program to make the comparison case-insensitive.

```
public class Q8_CaseInsensitive {  
    public static void main(String[] args) {  
        String str = "java";  
        if(str.equalsIgnoreCase("JAVA")){  
            System.out.println("Match");  
        }  
    }  
}
```

Your Code

```
1 import java.util.*;  
2 public class Q8_CaseInsensitive {  
3     public static void main(String[] args) {  
4         String str = "java";  
5         if (str.equalsIgnoreCase("JAVA"))  
6         {  
7             System.out.println("Match");  
8         }  
9     }  
10 }
```

Input

stdin input

Output

Match

QUESTIONS

Q1

Array Index Traversal
5 marks

Q2

Null Pointer Handling
5 marks

Q3

String Comparison
5 marks

Q4

While Loop Correction
5 marks

Q5

Reverse String Logic
5 marks

Q6

Array Sum Calculation
5 marks

Q7

String Character Access
5 marks

Q8

Case Inensitive Comparison



Q9 Array Declaration

5 marks

The program contains an error in array declaration and initialization. Identify the syntax issue and correct it.

```
import java.util.Arrays;

public class Q9_ArrayDeclaration {
    public static void main(String[] args) {
        int arr[] = {1,2,3};
        System.out.println(Arrays.toString(arr));
    }
}
```

Your Code

```
1 import java.util.*;
2 public class Q9_ArrayDeclaration {
3     public static void main(String[] args)
4     {
5         int[] arr = {1, 2, 3};
6         System.out.println(Arrays.toString(arr));
7     }
8 }
```

Input

Output

```
[1, 2, 3]
```

Array Index Traversal
5 marks

Q2
Null Pointer Handling
5 marks

Q3
String Comparison
5 marks

Q4
While Loop Correction
5 marks

Q5
Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q8
Case Insensitive Comparison
5 marks

Q9
Array Declaration

A switch statement prints multiple outputs due to missing statements. Identify the problem and correct the code.

```
public class Q10_Switch {  
    public static void main(String[] args) {  
        int num = 2;  
        switch(num){  
            case 1: System.out.println("One"); break;  
            case 2: System.out.println("Two"); break;  
            case 3: System.out.println("Three"); break;  
        }  
    }  
}
```

Your Code

```
1 import java.util.*;  
2 public class Q10_Switch {  
3     public static void main(String[] args) {  
4         int num = 2;  
5         switch (num) {  
6             case 1: System.out.println("One");  
7                 break;  
8             case 2: System.out.println("Two");  
9                 break;  
10            case 3: System.out.println("Three");  
11                break;  
12        }  
13    }  
14 }
```

Input

stdin input

Output

Two

Visible Test Cases